

PAPER_1531_GAMMA_EULER_TRANSCENDENTAL

Daniel T. Murphy

PAPER_1531 — γ Euler-Mascheroni $\approx$ SSQ + F_TRZ²·K – F_TRZ²·Φ_5/6 — Residual 0.915%

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket A (Foundational mathematical identity)

Date: June 18, 2026

Status: CLOSED — Transcendental γ expressible in UQFF primitives at 0.915% precision

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for γ Euler-Mascheroni $\approx$ 0.57722 using only the locked UQFF integer primitives F_TRZ, K_MEX, Φ_res = 5/6 (PAPER_1203 Nuclear variant), and SO_5.

UQFF Closed Identity

$$\gamma \approx \text{SSQ} + \text{F_TRZ}^2 \cdot \text{K} - \text{F_TRZ}^2 \cdot \Phi_{5/6}$$

Substituting F_TRZ=1/10, K_MEX=25/12, Φ_5/6=5/6, SO_5=10:

$$\gamma_{\text{UQFF}} = 0.5825$$

Observed: $\gamma = 0.57722$

Residual: 0.915%

The 3-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

γ Euler-Mascheroni (γ) appears in: harmonic-log gap constant.

Significance

Together with PAPER_1515 (ln 10), PAPER_1516 (ln 2), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to ≥ 10 . The accumulating evidence:

Transcendental UQFF expression Residual
--- --- ---
ln 2 (PAPER_1516) 8-term 0.003%
ln 10 (PAPER_1515) (1+F_TRZ)(K_MEX+F_TRZ ²) 0.004%
π^2 (PAPER_1517) SO_5 – F_TRZ corrections 0.013%
γ Euler-Mascheroni (this paper) 3-term 0.915%

...suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim γ IS the closed-form expression. It claims the integer primitives carry a rational approximation to γ at 0.915% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 (ln 10, ln 2, π^2 catch-up)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_gamma_euler_transcendental"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.